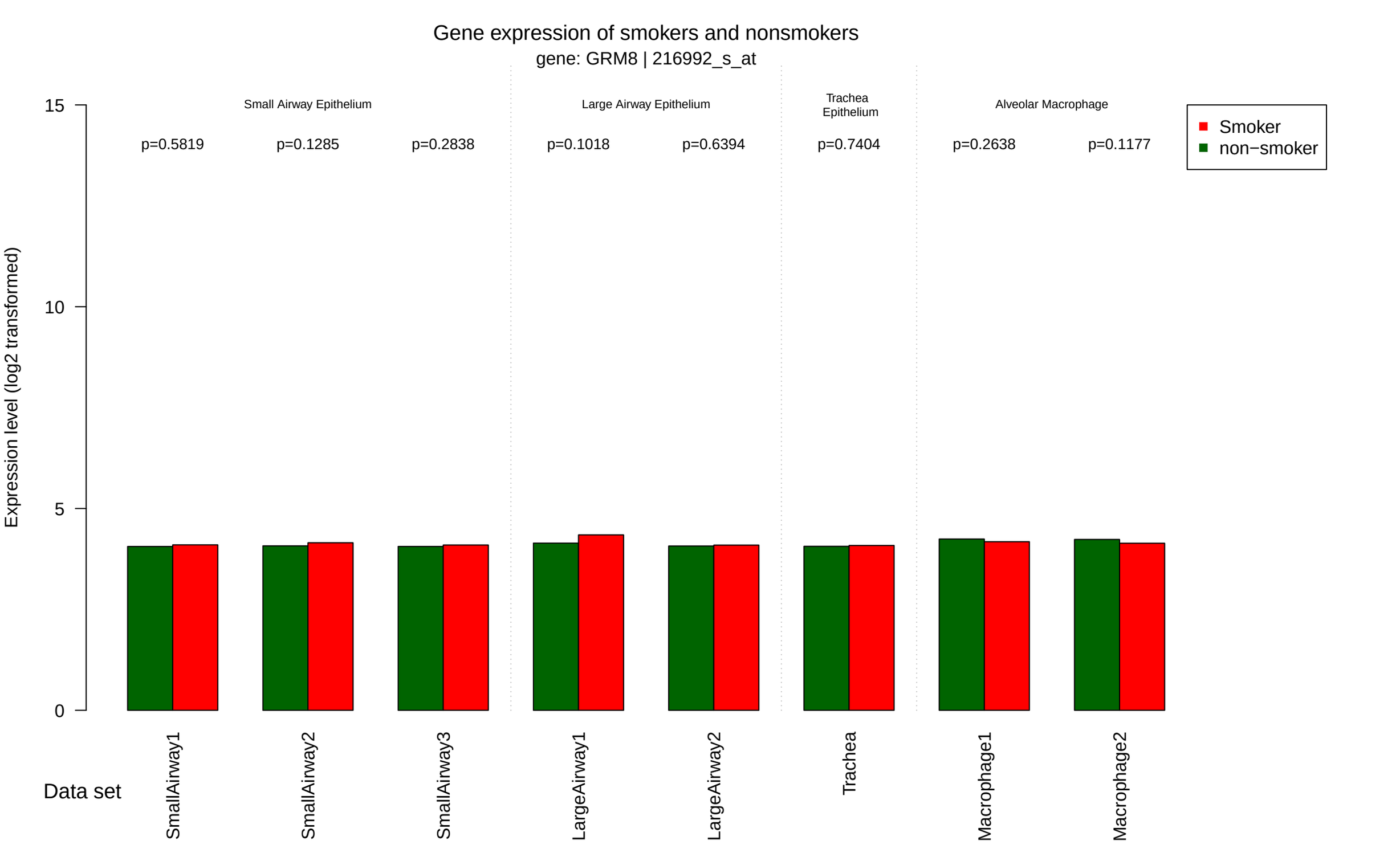
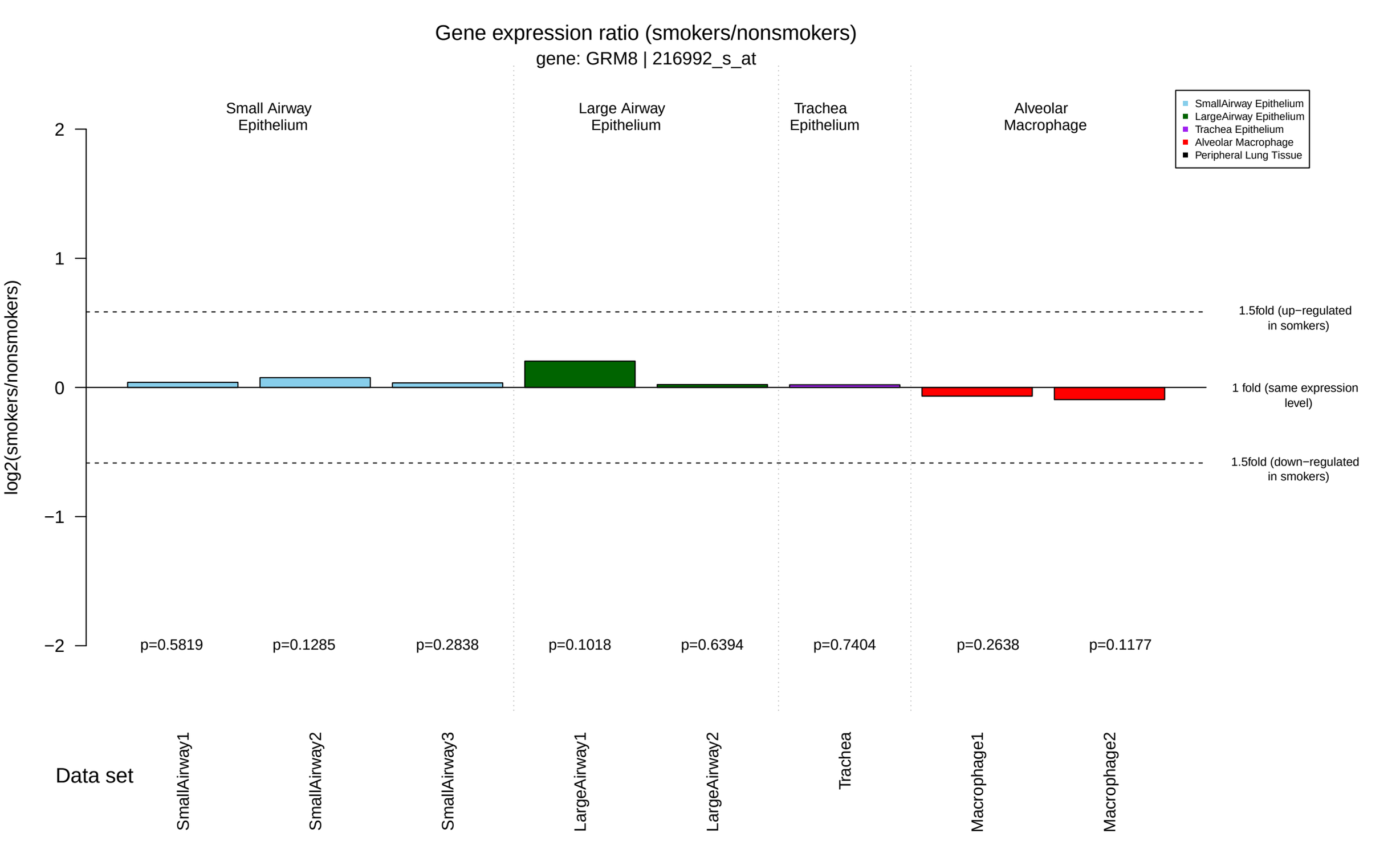
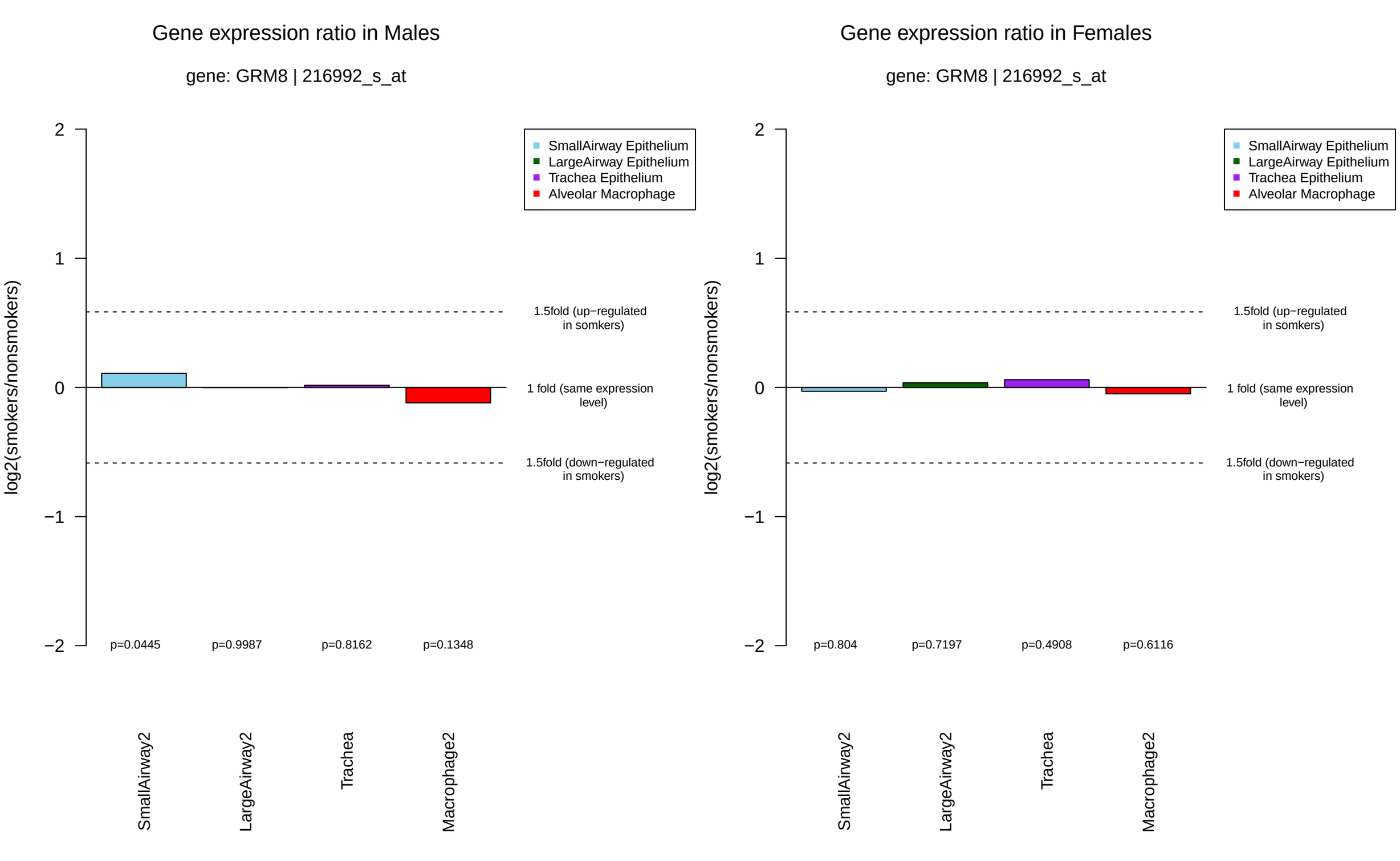
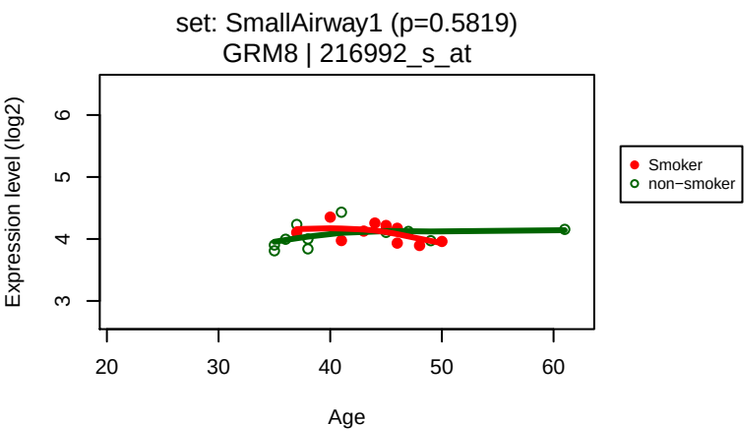
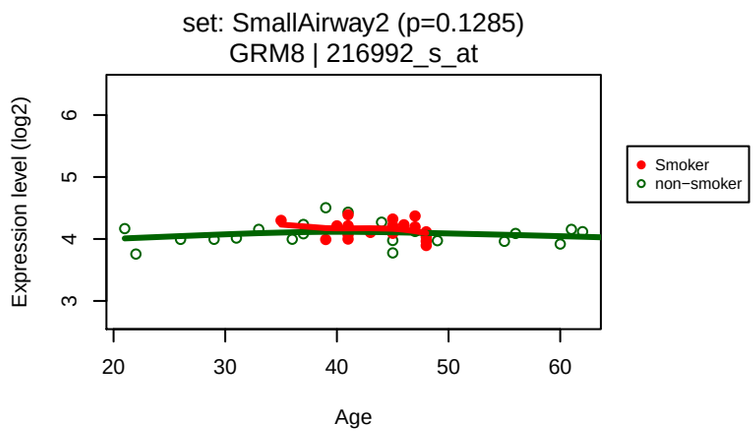
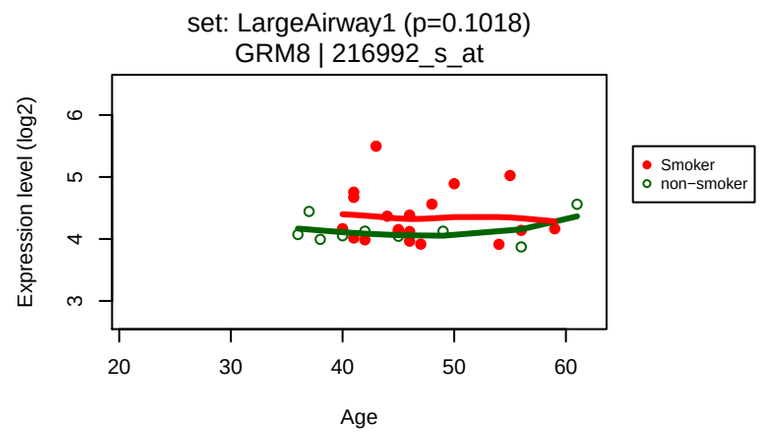

Gene expression of smokers and nonsmokers
gene: GRM8 | 216992_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5819
p=0.1285
p=0.2838
p=0.1018
p=0.6394
p=0.7404
p=0.2638
p=0.1177
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

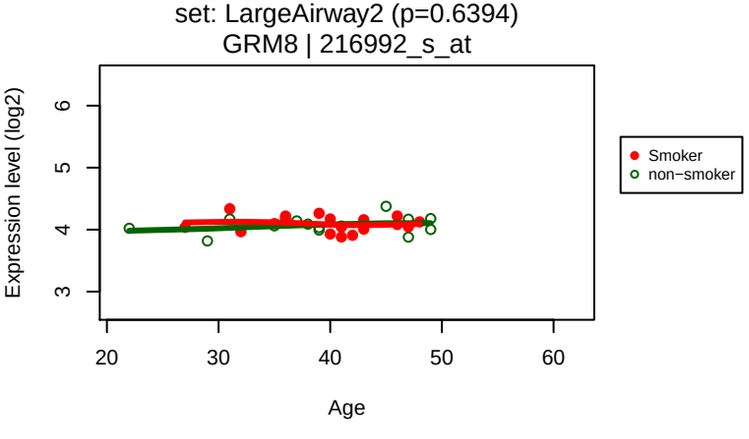

Gene expression ratio (smokers/nonsmokers)
gene: GRM8 | 216992_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5819
p=0.1285
p=0.2838
p=0.1018
p=0.6394
p=0.7404
p=0.2638
p=0.1177
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

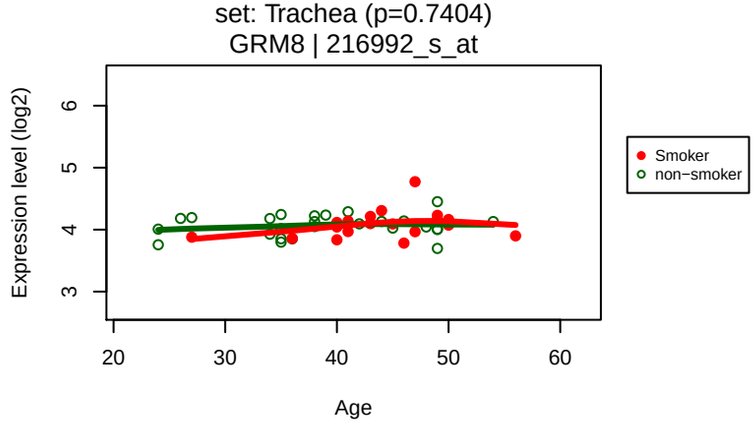

Gene expression ratio in Males
gene: GRM8 | 216992_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0445
p=0.9987
p=0.8162
p=0.1348
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRM8 | 216992_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.804
p=0.7197
p=0.4908
p=0.6116
SmallAirway2
LargeAirway2
Trachea
Macrophage2

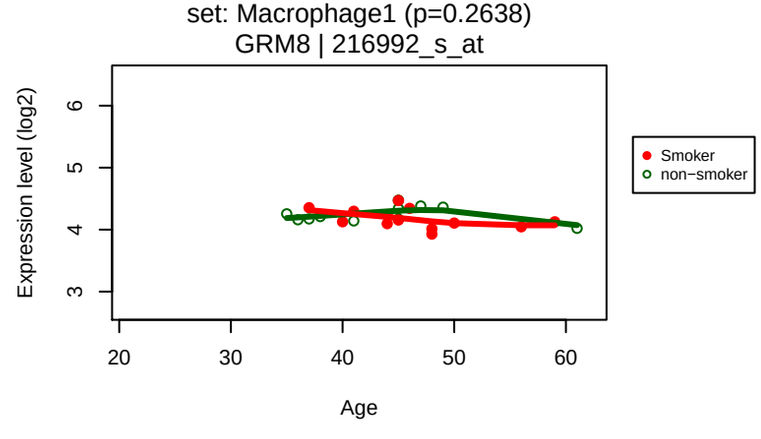

set: SmallAirway1 (p=0.5819)
GRM8 | 216992_s_at
Expression level (log2)
Age
Smoker
non−smoker

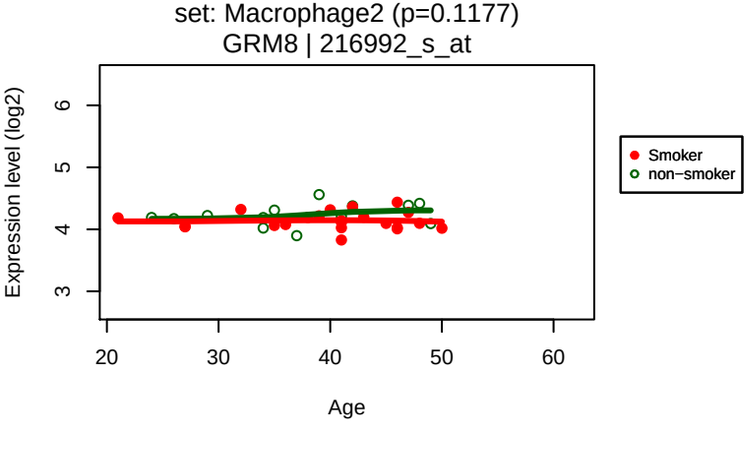

set: SmallAirway2 (p=0.1285)
GRM8 | 216992_s_at
Expression level (log2)
Age
Smoker
non−smoker

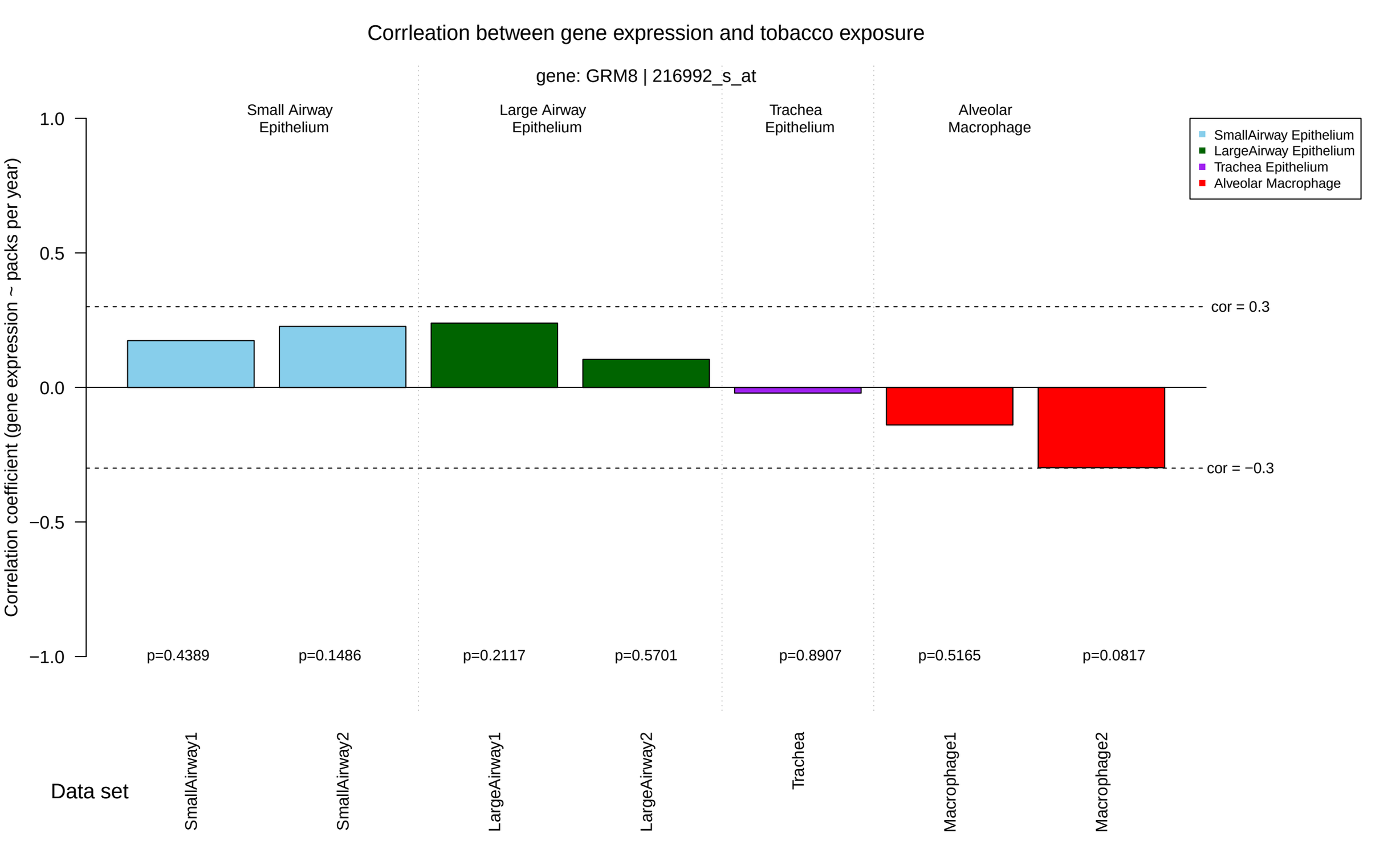

set: LargeAirway1 (p=0.1018)
GRM8 | 216992_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6394)
GRM8 | 216992_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7404)
GRM8 | 216992_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2638)
GRM8 | 216992_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1177)
GRM8 | 216992_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRM8 | 216992_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4389
p=0.1486
p=0.2117
p=0.5701
p=0.8907
p=0.5165
p=0.0817
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2